Abu Henaf Rashid Ahmad Mufti

Machine Learning Researcher | Deep Learning | Explainable AI | Computer Vision

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Final-year Computer Science student with research experience in machine learning, deep learning, computer vision, explainable AI, and predictive analytics. Experienced in dataset curation, image and video annotation using Roboflow, OCR dataset preparation, and YOLO-based object detection workflows. Passionate about building efficient, data-driven AI solutions for real-world applications.

EDUCATION

United International University — BSc in Computer Science and Engineering *Expected Grad. Early 2027*

Academia School, Mirpur, Dhaka — A Level & O Level, Edexcel Curriculum (Edexcel Award, Daily Star Award)

RESEARCH INTERESTS

Machine Learning, Deep Learning, Explainable AI, Computer Vision, Medical Image Analysis, Biomedical Machine Learning, NLP for Low-Resource Languages

SKILLS

Machine Learning: Scikit-learn, XGBoost, Random Forest, SVM

Deep Learning: TensorFlow, PyTorch, CNNs, Transfer Learning

Computer Vision: OpenCV, YOLO, Roboflow, Data Annotation, Dataset Curation

Explainable AI: SHAP, Feature Importance

Generative AI: LangChain, LangGraph, RAG, ChromaDB

Programming: Python, SQL, C++, Git, Docker

PUBLICATIONS

An Explainable and Scalable Framework for Compost Maturity Prediction (ACM Digital Library) *2024*

Peer-reviewed conference paper

- Developed an explainable machine learning framework for compost maturity prediction using real-world environmental data.
- Engineered physical, chemical, and biological features to improve prediction robustness.
- Evaluated ensemble learning models and explainability techniques for interpretable decision-making.

RESEARCH

A Multimodal Risk Assessment and Monitoring Framework for Assisted Elderly Living *2026 – Present*

Final Year Design Project (FYDP)

- Building a signal-based ensemble classifier that fuses physiological and behavioural sensor data to flag health risks in assisted-living settings.
- Annotated elderly behaviour videos using Roboflow to prepare datasets for YOLO-based behaviour detection models.
- Assisted in dataset curation and annotation quality control for computer vision model development.
- Designing the alerting layer that surfaces detected risks to caregivers in real time.

DNA Protein-Site Binding Evaluation *Oct 2025 – Finalised*

Machine Learning Project

- Designed machine learning pipelines for protein-binding prediction using high-dimensional biosensor data.
- Engineered structured features from biological signal curves for interpretable predictive modelling.

Pneumonia Lung Disease Classification *Oct 2025 – Finalised*

Digital Image Processing Project

– Developed a resource-efficient pneumonia classification framework for low-compute environments.
– Applied image preprocessing, segmentation, feature engineering, and deep learning using TensorFlow and PyTorch.
– Investigated lightweight alternatives to GPU-intensive inference for deployment on commodity hardware.

AQI Prediction and Generalisation *2026 – Present*

Peer Collaboration

– Compared Random Forest, XGBoost, CatBoost, and ensemble methods for AQI prediction.
– Applied feature importance analysis using SHAP to improve interpretability and model understanding.

Shadhu Bangla OCR and NLP Framework *2026 – Present*

Data Mining Research Project

– Built and managed annotated OCR datasets using Roboflow for historical Bangla manuscripts.
– Performed image annotation, dataset versioning, augmentation, and quality control to improve OCR model performance.
– Developed transformer-based OCR pipelines for low-resource scripts.

COMPETITIONS

University Physics Competition (UPC) 2025

Accomplished Competitor — Ideation and Methodology

APPLIED PROJECTS

Unified MediChat

– Designed an AI healthcare assistant combining retrieval-augmented generation, web and academic search, and multimodal input for evidence-informed responses.

Research Assist

– Built a research automation tool that retrieves, analyses, and synthesises academic literature into structured summaries.

Data Seeker

– Developed a natural language to SQL pipeline with query validation and automated analytical reporting.

CERTIFICATIONS

HackerRank — SQL (Advanced)

freeCodeCamp — Machine Learning with Python

freeCodeCamp — Data Analysis with Python

Anthropic Education — AI Capabilities and Limitations

Anthropic Education — AI Fluency: Framework & Foundations